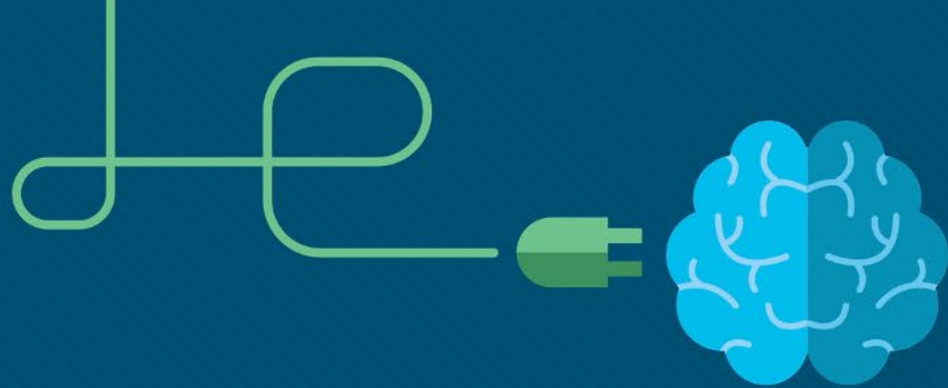




Module 3: Number Systems

Networking Devices and Initial Configuration
(NBIC)



3.1 Binary Number System

Dezimalsystem

Zahlenvorrat: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
Basis: 10

Basis	10	10	10	10
Position in Zahl	3	2	1	0
Berechnen	(10^3)	(10^2)	(10^1)	(10^0)
Stellenwert	1000	100	10	1

	Tausender	Hunderter	Zehner	Einer
Stellenwert	1000	100	10	1
Dezimalzahl (1473)	1	4	7	3
Berechnen	1 x 1000	4 x 100	7 x 10	3 x 1
Addiere...	1.000	+ 400	+ 70	+ 3
Ergebnis	1.473			

Binärsystem

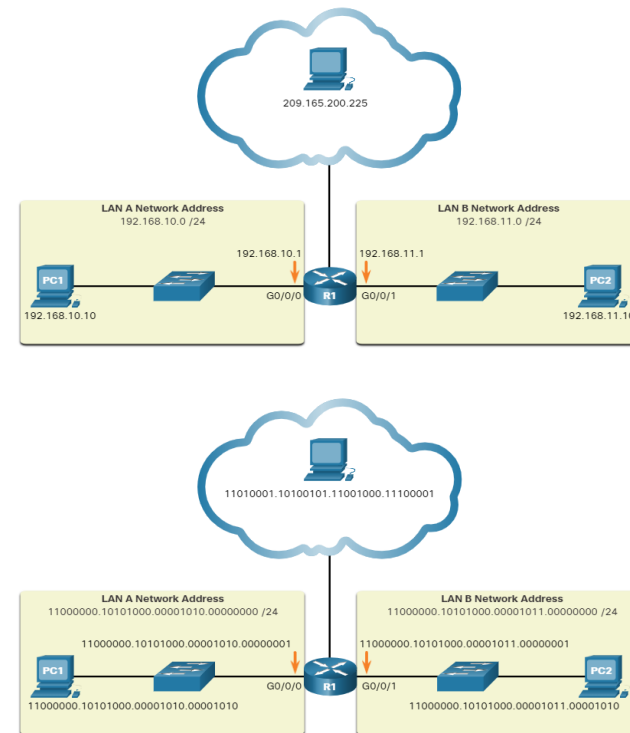
Zahlenvorrat: 0,1
Basis: 2

Basis	2	2	2	2	2	2	2	2
Position in der Zahl	7	6	5	4	3	2	1	0
Berechnen	(2^7)	(2^6)	(2^5))	(2^4))	(2^3)	(2^2)	(2^1)	(2^0)
Stellenwert	128	64	32	16	8	4	2	1

Stellenwert	128	64	32	16	8	4	2	1
Binäre Zahl (11000000)	1	1	0	0	0	0	0	0
Berechnen	1x128	1x64	0x32	0x16	0x8	0x4	0x2	0x1
Addiere...	128	+ 64	+ 0	+ 0	+ 0	+ 0	+ 0	+ 0
Ergebnis	192							

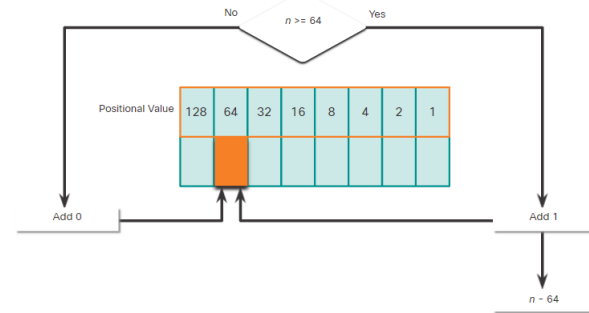
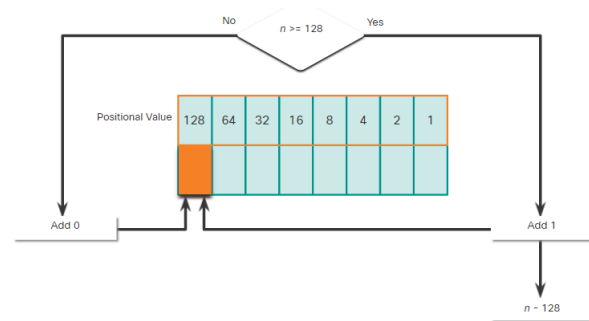
Binary and IPv4 Addresses

- IPv4 addresses are typically displayed using the decimal numbering system that consists of 10 digits which include 0 through 9.
 - The figure displays a sample topology using the dotted decimal numbering format to display IP addresses.
- However, hosts, servers, and network devices identify each other using the binary numbering system that consists of the digits 0 and 1, called bits.
- Network administrators must know binary addressing and gain practical skills to convert base two (binary) and base 10 (decimal) numbering systems.



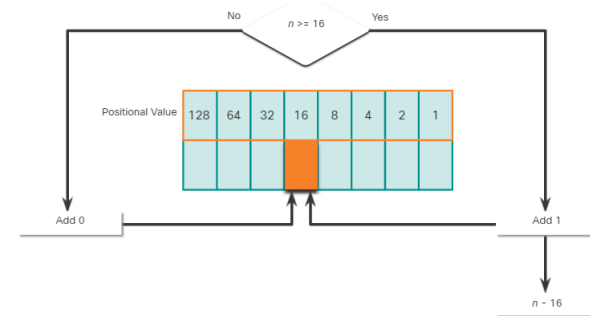
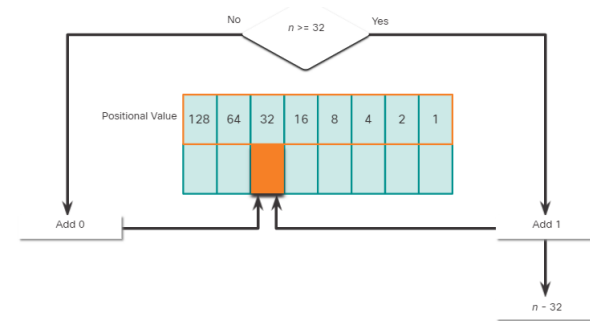
Decimal to Binary Conversion

- The binary positional value table is a valuable tool for converting a dotted-decimal IPv4 address to binary.
- Is the decimal number of the octet (n) equal to or greater than the most-significant bit (**128**)?
 - If no, then enter binary **0** in the **128** positional value.
 - If yes, add a binary **1** in the **128** positional value and subtract **128** from the decimal number.
- Is the decimal number of the octet (n) equal to or greater than the next most-significant bit (**64**)?
 - If no, then enter binary **0** in the **64** positional value.
 - If yes, add a binary **1** in the **64** positional value and subtract **64** from the decimal number.



Decimal to Binary Conversion (Cont.)

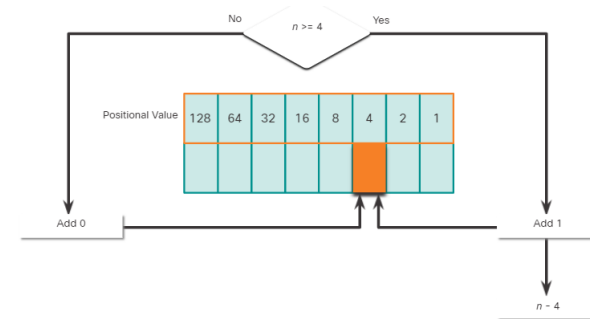
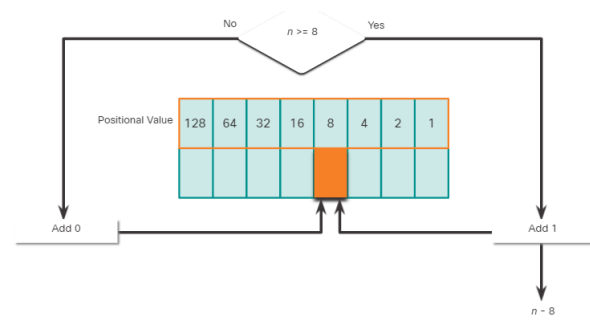
- Is the decimal number of the octet (n) equal to or greater than the most-significant bit (**32**)?
 - If no, then enter binary **0** in the **32** positional value.
 - If yes, add a binary **1** in the **32** positional value and subtract **32** from the decimal number.
- Is the decimal number of the octet (n) equal to or greater than the next most-significant bit (**16**)?
 - If no, then enter binary **0** in the **16** positional value.
 - If yes, add a binary **1** in the **16** positional value and subtract **16** from the decimal number.



Decimal to Binary Conversion (Cont.)

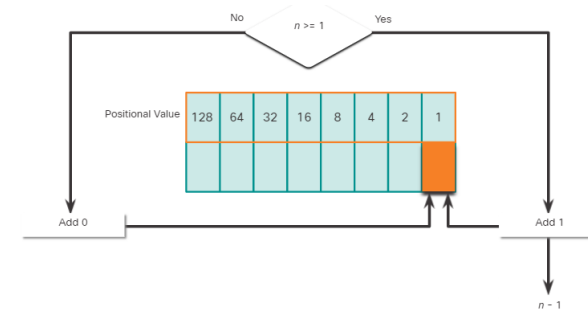
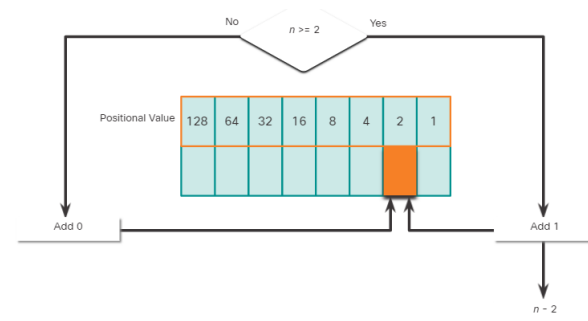
- Is the decimal number of the octet (n) equal to or greater than the most-significant bit (**8**)?
 - If no, then enter binary **0** in the **8** positional value.
 - If yes, add a binary **1** in the **8** positional value and subtract **8** from the decimal number.

- Is the decimal number of the octet (n) equal to or greater than the next most-significant bit (**4**)?
 - If no, then enter binary **0** in the **4** positional value.
 - If yes, add a binary **1** in the **4** positional value and subtract **4** from the decimal number.



Decimal to Binary Conversion (Cont.)

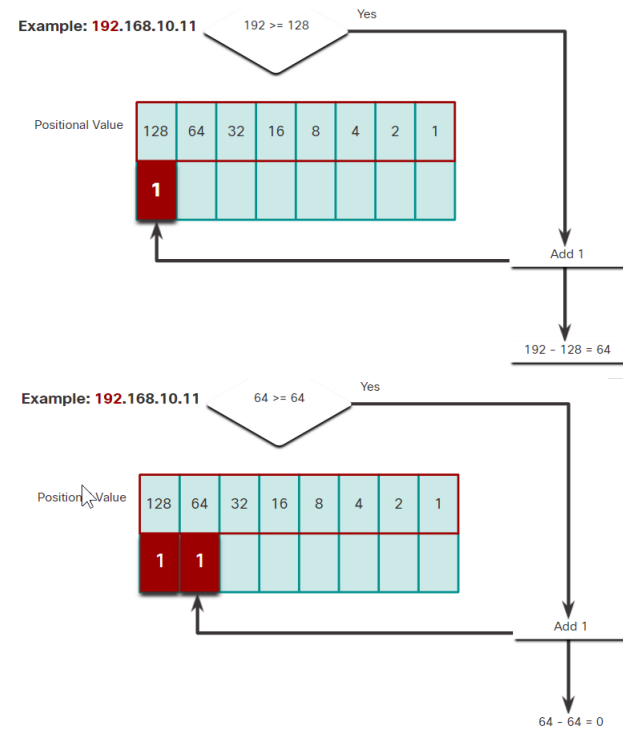
- Is the decimal number of the octet (**n**) equal to or greater than the most-significant bit (**2**)?
 - If no, then enter binary **0** in the **2** positional value.
 - If yes, add a binary **1** in the **2** positional value and subtract **2** from the decimal number.
- Is the decimal number of the octet (**n**) equal to or greater than the next most-significant bit (**1**)?
 - If no, then enter binary **0** in the **1** positional value.
 - If yes, add a binary **1** in the **1** positional value and subtract **1** from the decimal number.



Decimal to Binary Conversion Example

- To help understand the process, consider the IP address **192.168.11.10**.

- Is the first octet number **192** equal to or greater than the high-order bit **128**?
 - Yes, it is. Therefore, add a **1** to the high-order positional value to represent **128**.
 - Subtract **128** from **192** to produce a remainder of **64**.
- Is the remainder **64** equal to or greater than the next high-order bit **64**?
 - It is equal. Therefore add a **1** to the next high-order positional value.



Decimal to Binary Conversion Example (Cont.)

- Since there is no remainder, enter binary **0** in the remaining positional values.

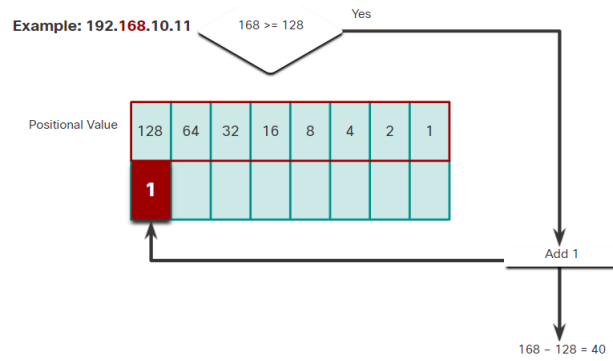
- The binary value of the first octet is **11000000**.

Example: **192.168.10.11**

Positional Value	128	64	32	16	8	4	2	1
	1	1	0	0	0	0	0	0

11000000

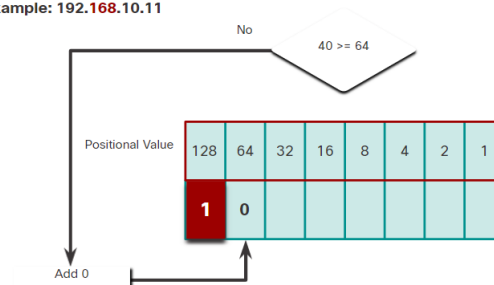
- Is the second octet number **168** equal to or greater than the high-order bit **128**?
 - It is, therefore, add a **1** to the high-order positional value to represent **128**.
 - Subtract **128** from **168** to produce a remainder of **40**.



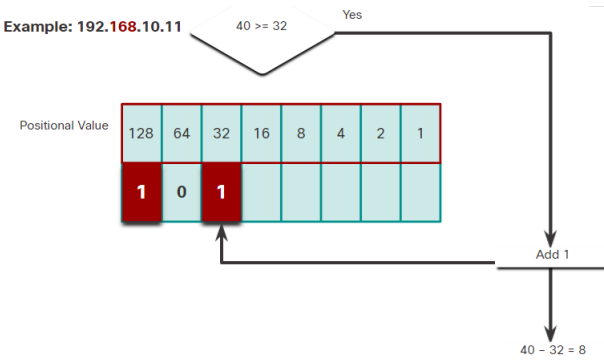
Decimal to Binary Conversion Example (Cont.)

- Is the remainder **40** equal to or greater than the next high-order bit **64**?
 - No, it is not, therefore, enter a binary **0** in the positional value.
- Is the remainder **64** equal to or greater than the next high-order bit **32**?
 - Yes, it is, therefore add a **1** to the high-order positional value to represent **32**.
 - Subtract **32** from **40** to produce a remainder of **8**.

Example: 192.168.10.11



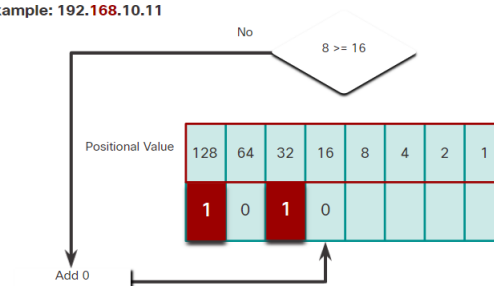
Example: 192.168.10.11



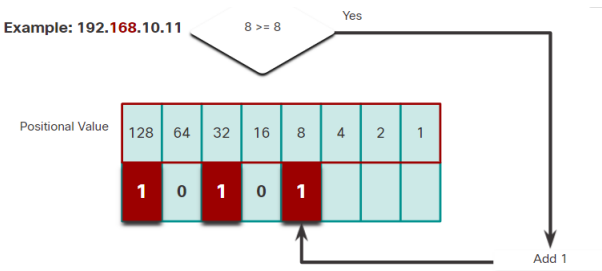
Decimal to Binary Conversion Example (Cont.)

- Is the remainder **8** equal to or greater than the next high-order bit **16**?
 - No, it is not, therefore, enter a binary **0** in the positional value.
- Is the remainder **8** equal to or greater than the next high-order bit **8**?
 - It is equal, therefore add a **1** to the next high-order positional value.

Example: 192.168.10.11



Example: 192.168.10.11



Decimal to Binary Conversion Example (Cont.)

- Since there is no remainder, enter binary **0** in the remaining positional values.
 - The binary value of the second octet is **10101000**.
- The binary value of the third octet is **00001010**.
- The binary value of the fourth octet is **00001011**.

Example: 192.**168**.10.11

Positional Value	128	64	32	16	8	4	2	1
	1	0	1	0	1	0	0	0

11000000 . **10101000** . . .

Example: 192.168.**10**.11

Positional Value	128	64	32	16	8	4	2	1
	0	0	0	0	1	0	1	0

11000000 . 10101000 . **00001010** . .

Example: 192.168.10.**11**

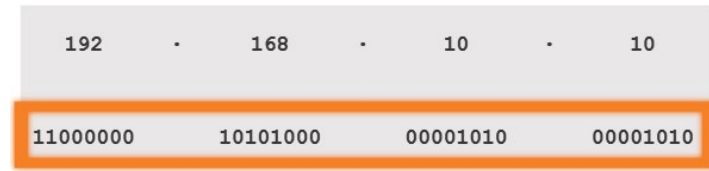
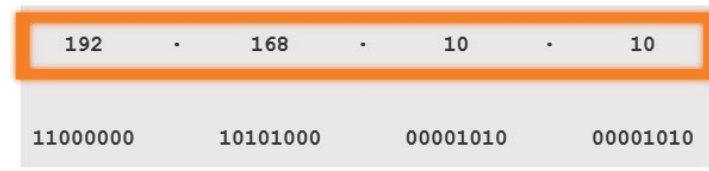
Positional Value	128	64	32	16	8	4	2	1
	0	0	0	0	1	0	1	1

11000000 . 10101000 . 00001010 . **00001011**

Binary Number System

IPv4 Addresses

- IPv4 addresses are typically displayed in dotted decimal format. It is assigned to a computer.
- **Dotted Decimal Address:**
 - 192.168.10.10 is an IP address assigned to a host.
- **4 Octets:**
 - The 192.168.10.10 address consists of four octets.
- **32-bit Address:**
 - The host stores the address as a 32-bit data stream.



3.2 Hexadecimal Number System

Hexadecimal and IPv6 Addresses

- IPv6 addresses use the hexadecimal base sixteen numbering system.
 - Hexadecimal uses 0 to 9 and letters A to F.
- To understand IPv6 addresses, you must be able to convert hexadecimal to decimal and vice versa.
- Binary and hexadecimal work well together because it is easier to express a value as a single hexadecimal digit than as four binary bits.
- The figure shows the equivalent decimal, binary, and hexadecimal values.

Decimal	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

Hexadezimalsystem

Zahlenvorrat: 0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F
Basis: 16

Basis	16	16	16	16
Position in Zahl	3	2	1	0
Berechnen	(16^3)	(16^2)	(16^1)	(16^0)
Stellenwert	4096	256	16	1

	Tausender	Hunderter	Zehner	Einer
Stellenwert	4096	256	16	1
Hexadezimal	1	B	7	A
Berechnen	1 x 4096	11 x 256	7 x 16	10 x 1
Addiere...	4096	+ 2816	+ 112	+ 10
Ergebnis	7034			

Hexadecimal Number System

Hexadecimal and IPv6 Addresses (Cont.)

- IPv6 and Ethernet MAC addresses use the hexadecimal numbering system to represent them.
- IPv6 addresses are 128 bits in length, and a single hexadecimal digit represents every 4 bits for a total of 32 hexadecimal values.
 - IPv6 addresses are not case-sensitive and can write in either lowercase or uppercase.
- Using a x:x:x:x:x:x:x:x format with each “x” consisting of four hexadecimal values, often referred to as a hextet, is how to write IPv6 addresses.
 - Each “x” is a single hextet, 16 bits, or four hexadecimal digits.

